Week 3 – Self-Attention:

How Transformers Decide Which Previous Words Matter

Order → Shipment → Receive → Restock → Inventory → Forecast

INVENTORY

1 Why Hidden Layers are Not Enough

Forecast → Embedding → Hidden Layer → Prediction

Explanation
Week 2 looked at only one word at a time. It ignores all the previous words.

Analogy
Reading just the last word of a sentence is not enough to understand the meaning.

Animation Idea: Only one word flows through the pipeline.

Summary:
- Single-word view = No Context

2 Problem Statement

Receive → Receive Shipment / Receive Payment

Explanation
The same word "Receive" can mean different things depending on context.

Analogy
Like the word "charge" in a restaurant bill vs a phone charger.

Animation Idea: Two different meanings pop up from the same word.

Summary:
- Words are ambiguous without context.

3 Introducing Context

Order · Shipment · Receive · Restock

Explanation
When predicting the next word after "Restock", the model should look at ALL previous words.

Analogy
Before making a decision, a manager reviews all previous steps.

Animation Idea: Arrows from Restock light up to all previous words.

Summary:
- Every previous word can be important.

4 Embedding Lookup

Order → ●●●●●●
Shipment → ●●●●●●
Receive → ●●●●●●
Restock → ●●●●●●
Embedding Vectors

Explanation
Each word is converted into a numerical vector that captures its meaning.

Analogy
Each word gets a unique fingerprint.

Animation Idea: Vectors build in with colors for each word.

Summary:
- Words → Vectors (Meaning Representation)

5 Creating Query, Key, Value

Word	Query (Ask)	Key (Describe)	Value (Contribute)
Order	●●●●●	●●●●●	●●●●●
Shipment	●●●●●	●●●●●	●●●●●
Receive	●●●●●	●●●●●	●●●●●
Restock	●●●●●	●●●●●	●●●●●

Explanation
From each embedding, we create three representations: Query, Key, Value.
- Query asks "What information am I looking for?"
- Key says "What information do I contain?"
- Value says "What information should I contribute?"

Animation Idea: Three colored vectors pop out from each embedding.

Summary:
- Each word has 3 roles: Ask, Describe, Contribute

6 Dot Product Attention

Restock (Query)

Order (Key)	Shipment (Key)	Receive (Key)	Restock (Key)
Score 1.2	Score 2.1	Score 3.4	Score 2.6

Explanation
The Query of "Restock" compares with every Key to measure similarity.

Analogy
Like asking a question and comparing it with every answer in a book.

Animation Idea: Blue query line moves across keys, scores appear.

Summary:
- Similarity scores decide importance.

7 Scaled Attention

Raw Scores: 1.2 · 2.1 · 3.4 · 2.6
↓
Divide by $\sqrt{d_k}$ (Scaler)
↓
Scaled Scores: 0.6 · 1.05 · 1.7 · 1.3

Explanation
Large scores can be unstable. We scale them down to keep them balanced.

Analogy
Like adjusting loud microphone volume before recording.

Animation Idea: Scores shrink smoothly after scaling.

Summary:
- Scaling keeps attention stable and fair.

8 Softmax

Scaled Scores: 0.6 · 1.05 · 1.7 · 1.3
↓
Softmax
↓
Attention Weights: 8% · 20% · 43% · 29%

Order · Shipment · Receive · Restock

Explanation
Softmax turns scores into probabilities that add up to 100%.

Analogy
Like converting exam marks into percentages.

Animation Idea: Bars grow to show final attention weights.

Summary:
Higher score → Higher attention weight

9 Attention Heatmap

Query	Keys: Order	Shipment	Receive	Restock
Restock (Query)	8%	20%	43%	29%

Low Attention — High Attention

Animation Idea: Heatmap intensifies from light → dark.

Summary:
Heatmap shows what the model focuses on.

10 Context Vector

Attention Weight		Value
8%	×	●●●●●
20%	×	●●●●●
43%	×	●●●●●
29%	×	●●●●●

→ Weighted Sum ●●●●●● → Context Vector

Animation Idea: Arrows flow into one vector that grows.

Summary:
Weighted combination of all Values.

11 Predicting Next Word

Context Vector ●●●●●● → Output Layer → Softmax

Attention Probabilities

Inventory	62%
Forecast	20%
Scenario	10%
Shipment	5%
Order	3%

Explanation
The context vector summarizes important information and helps predict the next word.

Analogy
Like summarizing the meeting and predicting the next agenda item.

Animation Idea: Probability bars fill up, top word glows.

Summary:
Context → Prediction of next word.

12 Putting Everything Together

Input Sequence (Order, Shipment, Receive, Restock) → Embedding Lookup → Query Key Value → Dot Product → Scaled Attention

Softmax → Attention Weights → Context Vector → Output Layer → Softmax → Predicted Word (Inventory)

Animation Idea: Full pipeline animates step-by-step.

Summary:
Self-Attention helps the model focus on what matters.

Key Takeaway

Self-Attention lets the model look back at all previous words and decide which ones matter most when predicting the next word.

Why It Matters

This ability to focus on relevant words is what makes Transformers powerful, accurate, and context-aware.

Remember

Not all previous words are equal.
Self-Attention decides what matters most.